

OpenCitations Data Management Plan

Version 1

Description

This Data Management Plan describes the management of the main data, data models, and software components of the OpenCitations infrastructure.

It covers the OpenCitations Data Model(OCDM), OpenCitations Index, OpenCitations Meta, and the software used to ingest, process, validate, and manage data compliant with the OpenCitations Data Model.

The purpose of the plan is to document how these research objects are created, processed, stored, documented, accessed, shared, preserved, and reused, including their formats, identifiers, provenance, licensing, and data quality procedures.

The DMP is intended to provide a structured and machine-actionable description of the data management practices adopted within OpenCitations.

Funder

UNIBO

Grant

Open Science Course

Researchers

Silvio Peroni (orcid:0000-0003-0530-4305), David Shotton (orcid:0000-0001-5506-523X), Marilena Daquino (0000-0002-1113-

7550), Giovanni Colavizza (0000-0002-9806-084X), Behnam Ghavimi (0000-0002-4627-5371), Anne Lauscher (0000-0001-8590-9827), Philipp Mayr (0000-0002-6656-1658), Matteo Romanello (0000-0002-7406-6286), Philipp Zumstein (0000-0002-6485-9434), Ivan Heibi (orcid:0000-0001-5366-5194), Arcangelo Massari (orcid:0000-0002-8420-0696), Marta Soricetti (orcid:0009-0008-1466-7742), Elia Rizzetto (0009-0003-7161-9310), Arianna Moretti (orcid:0000-0001-5486-7070), Fabio Mariani (0000-0002-7382-0187), Simone Persiani

Organizations

Alma Mater Studiorum University of Bologna

1. Main Info

Title of DMP: [OpenCitations Data Management Plan](#)

Description:

This Data Management Plan describes the management of the main data, data models, and software components of the OpenCitations infrastructure.

It covers the OpenCitations Data Model(OCDM), OpenCitations Index, OpenCitations Meta, and the software used to ingest, process, validate, and manage data compliant with the OpenCitations Data Model.

The purpose of the plan is to document how these research objects are created, processed, stored, documented, accessed, shared, preserved, and reused, including their formats, identifiers, provenance, licensing, and data quality procedures.

The DMP is intended to provide a structured and machine-actionable description of the data management practices adopted within OpenCitations.

Researchers:

[Silvio Peroni \(orcid:0000-0003-0530-4305\)](#)

[David Shotton \(orcid:0000-0001-5506-523X\)](#)

[Marilena Daquino \(0000-0002-1113-7550\)](#)

[Giovanni Colavizza \(0000-0002-9806-084X\)](#)

[Behnam Ghavimi \(0000-0002-4627-5371\)](#)

[Anne Lauscher \(0000-0001-8590-9827\)](#)

[Philipp Mayr \(0000-0002-6656-1658\)](#)

[Matteo Romanello \(0000-0002-7406-6286\)](#)

[Philipp Zumstein \(0000-0002-6485-9434\)](#)

[Ivan Heibi \(orcid:0000-0001-5366-5194\)](#)

[Arcangelo Massari \(orcid:0000-0002-8420-0696\)](#)

[Marta Soricetti \(orcid:0009-0008-1466-7742\)](#)

[Elia Rizzetto \(0009-0003-7161-9310\)](#)

[Arianna Moretti \(orcid:0000-0001-5486-7070\)](#)

[Fabio Mariani \(0000-0002-7382-0187\)](#)

[Simone Persiani](#)

Organizations:

Alma Mater Studiorum University of Bologna

Contact: Wu Yifan (yifan.wu2@studio.unibo.it), Peroni Silvio (silvio.peroni@unibo.it)

2. Funding

Funding organizations: UNIBO

Grants: Open Science Course

Project:

3. License

License: CC-BY-4.0

Access Rights: Public

4. Templates

Descriptions

OpenCitations Data Model

The OpenCitations Data Model (OCDM) defines the RDF-based representation of bibliographic entities, citations, identifiers, agents, publication venues, and provenance in OpenCitations. It reuses established ontologies, including the SPAR Ontologies, to support consistent representation of bibliographic and citation data.

Template: Data Management Plan Template - University of Bologna

Type: Dataset

1 Your research output

1.1 Contact details and responsibilities

1.1.1 Contact person(s)

Contact Name	Contact ID	E-mail address	Affiliation
Peroni, Silvio	0000-0003-0530-4305	silvio.peroni@unibo.it	ALMA MATER STUDIORUM - Università di Bologna

1.1.3 Research output creator(s) and contributor(s)

Contributor Name	Contributor ID	E-mail address	Affiliation	Role
Daquino, Marilena	0000-0002-1113-7550	marilena.daquino2@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator
Shotton, David	0000-0001-5506-523X	david.shotton@opencitations.net	University of Oxford	Creator

1.1.4 Institution(s) involved

- a. ALMA MATER STUDIORUM - Università di Bologna
- b. University of Oxford

1.2 Resources for data management

1.2.1 What resources will be dedicated to data management and ensuring that data will be FAIR?

Resources supporting the FAIR management of OCDM include its public specification and documentation, version control through the OpenCitations GitHub repository, and versioned releases deposited in Figshare.

1.3 Identifying the research output

1.3.2 Which type of research output are you describing?

Dataset

2 Summary

2.1 Identification of the dataset

2.1.1 Are you re-using an already existing dataset?

Yes

2.1.2 Re-using an existing dataset: why and how

The project reuses the existing OpenCitations Data Model (OCDM) because it provides the common RDF-based model used across OpenCitations datasets and software.

OCDM documentation and the OpenCitations Ontology are released under CC BY 4.0, so reuse requires attribution.

2.2 Dataset characteristics

2.2.1 Dataset name

OpenCitations Data Model (OCDM)

2.2.2 Data types

- a. Citation data
- b. Bibliographic metadata
- c. Provenance data
- d. Structured semantic data

2.2.3 Data formats

- a. JSON-LD
- b. CSV

2.2.4 Size of data

other

2.2.5 How is the data generated/collected

OCDM was first developed in 2016 for the OpenCitations Corpus and has since been extended as new OpenCitations datasets and external use cases introduced additional requirements. Its development reuses established vocabularies and ontologies, including the SPAR Ontologies, PROV-O, Web Annotation, VoID, and DCAT. Extensions developed for the Open Biomedical Citations in Context Corpus followed the SAMOD methodology using motivating scenarios, competency questions, and a glossary of terms.

2.2.6 Why is the data generated/collected

OCDM provides a common model for bibliographic and citation data across sources that use different schemas, technologies, and vocabularies. This common representation reduces inconsistencies and supports data integration and reuse within OpenCitations and external projects.

2.2.7 Dataset description

The OpenCitations Data Model is an RDF-based model for representing bibliographic entities, citations, identifiers, agents, publication venues, and provenance. It reuses the SPAR Ontologies, PROV-O, and other established vocabularies to provide a consistent semantic representation across OpenCitations datasets and external projects.

3 Ethics and legal aspects

3.1 Personal data

3.1.1 Are you handling personal data?

No

3.2 Legal aspects

3.2.1 Are there any legal issues associated to your output? (e.g. IPR or valorization)

No

3.3 Other ethical aspects

3.3.1 Are there other ethical issues associated to your output?

No

4 Making research FAIR

4.1 Long-term preservation

4.1.1 Selecting what to preserve

The OCDM specification, ontology documentation, examples, development and evaluation materials, and previous published versions will be preserved. Earlier versions document changes to the model and support the interpretation of data created under older versions.

4.1.2 Chosen repository/ies

Name	Link	Is your repository OpenAIRE compliant?	Is your repository catalogued in re3data?	What type of repository is it?
Figshare	https://doi.org/10.6084/m9.figshare.3443876	Yes	Yes	Generalist

Comment:

The OCDM specification is deposited in Figshare under DOI 10.6084/m9.figshare.3443876. The OpenCitations GitHub repository is used for active maintenance and version control of the documentation and related development materials.

4.2 Findability

4.2.1 Persistent identifier(s)

Type of PID (e.g., DOI, Handle)	PID (e.g., doi.org/10...)
DOI	10.6084/m9.figshare.3443876.v7

4.2.2 Metadata

The Figshare record includes structured metadata for the title, creators, description, keywords, license, DOI, publication date, and version. Figshare exposes these metadata in machine-readable formats through its API and metadata export services.

4.2.3 Keywords

OpenCitations, OpenCitations Data Model, OCDM, bibliographic metadata, citation data, RDF, Semantic Web, SPAR Ontologies, provenance

4.3 Accessibility

4.3.1 Is the research output openly accessible?

Yes

4.3.3 Accessing the research output

The OCDM specification is available through its Figshare record under DOI 10.6084/m9.figshare.3443876. The OpenCitations GitHub repository provides the model documentation, development materials, and evaluation resources.

4.3.4 Is metadata openly accessible?

Yes

4.4 Interoperability

4.4.1 Methodologies

OCDM has evolved since its initial development in 2016 as new OpenCitations datasets and external use cases introduced additional requirements. It reuses established ontologies and vocabularies, and extensions for the Open Biomedical Citations in Context Corpus followed SAMOD using motivating scenarios, competency questions, and a glossary of terms. Human-readable documentation, JSON-LD examples, versioning, and provenance information support the interpretation and reuse of the model.

4.4.2 Vocabularies, taxonomies and other standards

OCDM uses RDF as its underlying representation model and reuses terms through the OpenCitations Ontology (OCO). OCO includes terms from the SPAR Ontologies, including CiTO, FaBiO, BiRO, C4O, DEO, and the DataCite Ontology. OCDM also uses PROV-O for provenance, the Web Annotation Ontology for annotations, VoID for dataset descriptions, and DCAT for dataset and distribution metadata. JSON-LD is used for RDF examples and machine-readable representations, while CSV is used for selected tabular representations.

4.5 Reusability

4.5.1 Licensing

CC BY 4.0, "Attribution", <https://creativecommons.org/licenses/by/4.0/>

Comment:

OCDM and its documentation are released under CC BY 4.0.

4.5.2 Documentation

OCDM documentation defines its terms, naming conventions, URI patterns, and data modelling rules, with real-world examples. Machine-readable examples are provided in JSON-LD, with CSV templates for selected tabular representations. Development materials include motivating scenarios, competency questions, a glossary of terms, and evaluation resources available through the OpenCitations GitHub repository.

5 Quality and security

5.1 Quality of the research output

5.1.1 Measures implemented to ensure quality

OCDM reuses established ontologies and follows a documented ontology development and evaluation process. Extensions use SAMOD with motivating scenarios, competency questions, and a glossary of terms. OCO has been evaluated against JATS/XML data, compared with other bibliographic data models, and tested against requirements from early adopters. New use cases are used to identify and address gaps in the model.

5.2 Security measures

5.2.1 Storage

Cloud Storage Services

Comment:

OCDM documentation and development materials are maintained in the public OpenCitations GitHub repository with Git version history. Published versions of the specification are deposited in Figshare with persistent DOIs.

5.2.2 Backup

Recovery relies on Git version history and versioned OCDM releases deposited in Figshare, which allow earlier states of the model and documentation to be restored.

OpenCitations Index data

The OpenCitations Index is an open dataset that integrates citation links from Crossref, the NIH Open Citation Collection, DataCite, OpenAIRE, and the Japan Link Center. Citations are deduplicated, represented according to the OpenCitations Data Model, and accompanied by provenance and change-tracking information.

Template: [Data Management Plan Template - University of Bologna](#)

Type: [Dataset](#)

1 Your research output

1.1 Contact details and responsibilities

1.1.1 Contact person(s)

Contact Name	Contact ID	E-mail address	Affiliation
Heibi, Ivan	0000-0001-5366-5194	ivan.heibi2@unibo.it	ALMA MATER STUDIORUM - Università di Bologna

1.1.3 Research output creator(s) and contributor(s)

Contributor Name	Contributor ID	E-mail address	Affiliation	Role
Moretti, Arianna	0000-0001-5486-7070	arianna.moretti4@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator

Soricetti, Marta	0009-0008-1466-7742	marta.soricetti2@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator
Peroni, Silvio	0000-0003-0530-4305	silvio.peroni@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Other (project administration, resources and supervision)

1.1.4 Institution(s) involved

ALMA MATER STUDIORUM - Università di Bologna

1.2 Resources for data management

1.2.1 What resources will be dedicated to data management and ensuring that data will be FAIR?

Resources supporting FAIR management of the OpenCitations Index include the ingestion workflow, provenance and change-tracking information, public data dumps, REST APIs and the SPARQL endpoint.

1.3 Identifying the research output

1.3.2 Which type of research output are you describing?

Dataset

2 Summary

2.1 Identification of the dataset

2.1.1 Are you re-using an already existing dataset?

Yes

2.1.2 Re-using an existing dataset: why and how

The OpenCitations Index is reused because it provides a unified, deduplicated collection of citation links integrated from multiple scholarly data sources.

OpenCitations ingests citation data from sources that provide the relevant data under CC0 or a compatible open license. OpenCitations Index distributions are released under CC0.

2.2 Dataset characteristics

2.2.1 Dataset name

OpenCitations Index

2.2.2 Data types

- a. Citation data
- b. Citation metadata
- c. Provenance data
- d. Change-tracking data
- e. Persistent identifier data

2.2.3 Data formats

- a. CSV
- b. N-Triples
- c. Scholix

2.2.4 Size of data

terabyte (TB)

Comment:

The July 2026 OpenCitations Index dump contains more than 2.5 billion citations. Depending on the distribution format, the uncompressed data range from hundreds of gigabytes to several terabytes.

2.2.5 How is the data generated/collected

Citation data are collected from Crossref, NIH-OCC, DataCite, OpenAIRE, and the Japan Link Center. During Source Preprocess, source data are converted into standardized citation tables. The Meta Process maps external identifiers to OpenCitations Meta Identifiers (OMIDs), creating new OMIDs when required. The

Index Process then converts citation links into OMID-to-OMID relationships, removes duplicates, and generates the OpenCitations Index data.

2.2.6 Why is the data generated/collected

The OpenCitations Index provides a unified collection of open citation data from multiple sources. Mapping external identifiers to OMIDs allows equivalent citation links to be recognised and deduplicated, supporting citation analysis and reuse by research tools and external services.

2.2.7 Dataset description

The OpenCitations Index is an open collection of citation links integrated from multiple scholarly data sources. Citations are normalised as OMID-to-OMID relationships, deduplicated, and represented according to the OpenCitations Data Model. Each citation is identified by an Open Citation Identifier (OCI) and can include metadata, provenance, and change-tracking information.

3 Ethics and legal aspects

3.1 Personal data

3.1.1 Are you handling personal data?

No

3.2 Legal aspects

3.2.1 Are there any legal issues associated to your output? (e.g. IPR or valorization)

No

3.3 Other ethical aspects

3.3.1 Are there other ethical issues associated to your output?

No

4 Making research FAIR

4.1 Long-term preservation

4.1.1 Selecting what to preserve

Citation data, provenance, change-tracking information, published distributions, and previous dataset dumps will be preserved. Earlier dumps record the state of the Index at different points in time and support reproducibility.

4.1.2 Chosen repository/ies

Name	Link	Is your repository OpenAIRE compliant?	Is your repository catalogued in re3data?	What type of repository is it?
Figshare	https://figshare.com	Yes	Yes	Generalist

Comment:

Published OpenCitations Index distributions and their provenance data are deposited in Figshare.

4.2 Findability

4.2.1 Persistent identifier(s)

Type of PID (e.g., DOI, Handle)	PID (e.g., doi.org/10...)
DOI	10.6084/m9.figshare.24356626
DOI	10.6084/m9.figshare.24369136
DOI	10.6084/m9.figshare.24416749

4.2.2 Metadata

Index citations include identifiers for the citing and cited resources, citation metadata where available, and provenance records describing their source and history. Figshare records provide descriptive metadata and persistent DOIs for the published distributions.

4.2.3 Keywords

OpenCitations, OpenCitations Index, citations, bibliographic metadata, RDF, Linked Open Data, provenance, OCI, OMID

4.3 Accessibility

4.3.1 Is the research output openly accessible?

Yes

4.3.3 Accessing the research output

The OpenCitations Index is available through downloadable dataset dumps, a SPARQL endpoint, REST APIs, and web interfaces. Published distributions can be downloaded from Figshare.

4.3.4 Is metadata openly accessible?

Yes

4.4 Interoperability

4.4.1 Methodologies

The OpenCitations Index ingestion workflow consists of Source Preprocess, Meta Process, and Index Process. Source Preprocess converts heterogeneous source data into standardized CSV citation tables. The Meta Process maps external identifiers to OMIDs, and the Index Process converts citation links into OMID-to-OMID relationships, removes duplicates, assigns OCIs, and generates the final distributions. Provenance and change history are generated alongside the citation data.

4.4.2 Vocabularies, taxonomies and other standards

The OpenCitations Index follows the OpenCitations Data Model and represents citation data in RDF. OMIDs identify the citing and cited bibliographic resources across different external identifier schemes, while each citation is assigned an Open Citation Identifier (OCI). PROV-O is used for provenance and versioning, and Scholix is supported as an exchange format for citation links.

4.5 Reusability

4.5.1 Licensing

CC 0, "No Rights Reserved", <https://creativecommons.org/share-your-work/public-domain/cc0/>

4.5.2 Documentation

The OpenCitations Index is documented in its canonical research paper and on the OpenCitations website. The documentation covers data sources, the ingestion workflow, citation modelling, identifiers, provenance, published distributions, and access services.

5 Quality and security

5.1 Quality of the research output

5.1.1 Measures implemented to ensure quality

The ingestion workflow normalises external identifiers and uses OpenCitations Meta to map identifiers referring to the same bibliographic resource. These mappings allow duplicate citation links from different sources to be identified and removed. Provenance and change-tracking records support traceability, while validation is applied before publication.

5.2 Security measures

5.2.1 Storage

Other

Comment:

The active OpenCitations Index is stored in a QLever RDF triplestore, while published dataset distributions are stored separately in Figshare.

5.2.2 Backup

Recovery is supported by published dataset dumps and provenance distributions in Zenodo, Figshare, and Internet Archive, with previous dumps.

OpenCitations Meta data

OpenCitations Meta is an open bibliographic metadata database for scholarly resources involved in citations indexed by OpenCitations. It integrates and curates metadata from external scholarly sources, assigns OpenCitations Meta Identifiers (OMIDs), and represents the curated records as RDF compliant with the OpenCitations Data Model. Provenance and change-tracking records document the source and history of each entity.

Template: [Data Management Plan Template - University of Bologna](#)

Type: [Dataset](#)

1 Your research output

1.1 Contact details and responsibilities

1.1.1 Contact person(s)

Contact Name	Contact ID	E-mail address	Affiliation
Massari, Arcangelo	0000-0002-8420-0696	arcangelo.massari@unibo.it	ALMA MATER STUDIORUM - Università di Bologna

1.1.3 Research output creator(s) and contributor(s)

Contributor Name	Contributor ID	E-mail address	Affiliation	Role
Mariani, Fabio	0000-0002-7382-0187	fabio.mariani@leuphana.de	Leuphana University Lüneburg	Creator

Peroni, Silvio	0000-0003-0530-4305	silvio.peroni@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Other (conceptualization, methodology, project administration and supervision)
Shotton, David	0000-0001-5506-523X	david.shotton@opencitations.net	University of Oxford	Other (conceptualization, project administration and supervision)
Heibi, Ivan	0000-0001-5366-5194	ivan.heibi2@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Other (writing review and editing)

1.1.4 Institution(s) involved

- a. ALMA MATER STUDIORUM - Università di Bologna
- b. Leuphana Universität Lüneburg
- c. University of Oxford

1.2 Resources for data management

1.2.1 What resources will be dedicated to data management and ensuring that data will be FAIR?

Resources supporting FAIR management of OpenCitations Meta include its automated curation and RDF-conversion workflow, provenance and change tracking, public dataset distributions, APIs and the SPARQL endpoint.

1.3 Identifying the research output

1.3.2 Which type of research output are you describing?

Dataset

2 Summary

2.1 Identification of the dataset

2.1.1 Are you re-using an already existing dataset?

Yes

2.1.2 Re-using an existing dataset: why and how

OpenCitations Meta is reused because it provides curated bibliographic metadata for scholarly resources involved in citations indexed by OpenCitations.

OpenCitations Meta data are released under CC0.

2.2 Dataset characteristics

2.2.1 Dataset name

OpenCitations Meta

2.2.2 Data types

- a. Bibliographic metadata
- b. Persistent identifier data
- c. Responsible-agent metadata
- d. Provenance and change-tracking data

2.2.3 Data formats

- a. CSV
- b. JSON-LD (RDF)

2.2.4 Size of data

gigabytes (GB)

Comment:

The June 2026 OpenCitations Meta release contains more than 147 million bibliographic entities and 422 million author roles. The CSV distribution is approximately 8.4 GB compressed and 56 GB uncompressed; the RDF metadata and provenance distribution is approximately 45.7 GB compressed and 72 GB uncompressed.

2.2.5 How is the data generated/collected

OpenCitations Meta ingests bibliographic metadata from scholarly sources including Crossref, DataCite, and PubMed, with additional sources incorporated in later releases. Input metadata are converted to a standard CSV structure and processed by the Curator for deduplication, enrichment, normalization, and error correction. Bibliographic entities are assigned OMIDs, after which the Creator converts the curated data to OCDM-compliant RDF and generates provenance and change-tracking records.

2.2.6 Why is the data generated/collected

OpenCitations Meta provides locally maintained bibliographic metadata for resources involved in OpenCitations citations, reducing dependence on real-time queries to external services. OMIDs map different external identifiers to the same bibliographic resource and provide identifiers for resources that lack an external persistent identifier.

2.2.7 Dataset description

OpenCitations Meta is an open bibliographic metadata database for scholarly resources involved in citations indexed by OpenCitations. Records are automatically curated and deduplicated using persistent identifiers, and entities are assigned OpenCitations Meta Identifiers (OMIDs). Curated metadata are represented as OCDM-compliant RDF, with provenance and change-tracking records documenting the source and history of each entity.

3 Ethics and legal aspects

3.1 Personal data

3.1.1 Are you handling personal data?

Yes

3.1.2 Solutions adopted for handling personal data

Personal data in OpenCitations Meta are limited to publicly available scholarly metadata such as author and editor names and ORCID identifiers obtained from external bibliographic sources. The dataset does not involve participant recruitment or direct collection from individuals. Provenance records retain the source of imported metadata and subsequent changes.

3.2 Legal aspects

3.2.1 Are there any legal issues associated to your output? (e.g. IPR or valorization)

No

3.3 Other ethical aspects

3.3.1 Are there other ethical issues associated to your output?

No

4 Making research FAIR

4.1 Long-term preservation

4.1.1 Selecting what to preserve

Curated bibliographic metadata, OMID mappings, provenance records, change-tracking information, published distributions, and previous releases will be preserved. Earlier releases document the state of the database at different points in time and support reproducibility and historical analysis.

4.1.2 Chosen repository/ies

Name	Link	Is your repository OpenAIRE compliant?	Is your repository catalogued in re3data?	What type of repository is it?
Zenodo	https://zenodo.org	Yes	Yes	Generalist
Figshare	https://figshare.com	Yes	Yes	Generalist

Comment:

Current OpenCitations Meta releases are deposited in Zenodo, while earlier versioned releases remain available through Figshare.

4.2 Findability

4.2.1 Persistent identifier(s)

Type of PID (e.g., DOI, Handle)	PID (e.g., doi.org/10...)
DOI	10.5281/zenodo.20965426
DOI	10.5281/zenodo.20965956

Comment:

Repository DOIs identify published distributions, while OMIDs identify individual bibliographic entities within OpenCitations Meta.

4.2.2 Metadata

Published distributions include repository metadata for the title, creators, description, publication date, license, version, and persistent identifiers. Within OpenCitations Meta, bibliographic records contain structured metadata and identifiers, while provenance records document their source and modification history.

4.2.3 Keywords

OpenCitations, OpenCitations Meta, bibliographic metadata, OMID, RDF, Linked Open Data, provenance, change tracking, data curation

4.3 Accessibility

4.3.1 Is the research output openly accessible?

Yes

4.3.3 Accessing the research output

OpenCitations Meta is available through downloadable dataset dumps, a public REST API, and the OpenCitations SPARQL endpoint. Current downloads include database, CSV metadata, and RDF metadata and provenance distributions.

4.3.4 Is metadata openly accessible?

Yes

4.4 Interoperability

4.4.1 Methodologies

The OpenCitations Meta workflow begins with bibliographic metadata in a standardized eleven-column CSV structure. The Curator deduplicates, enriches, normalizes, and corrects records using persistent identifiers, and each entity is assigned an OMID. The Creator converts the curated data to OCDM-compliant RDF. Curated CSV and RDF are stored as files and used to populate the graph database, while provenance and change-tracking records document entity creation and subsequent modifications.

4.4.2 Vocabularies, taxonomies and other standards

OpenCitations Meta follows the OpenCitations Data Model and represents semantic data in RDF. PROV-O is used for provenance and change tracking. OMIDs identify entities within OpenCitations Meta, while external identifier schemes include DOI, PMID, ISBN, ISSN, and ORCID.

4.5 Reusability

4.5.1 Licensing

CC 0, "No Rights Reserved", <https://creativecommons.org/share-your-work/public-domain/cc0/>

4.5.2 Documentation

OpenCitations Meta is documented in its canonical research paper, the OpenCitations documentation and API pages, and the public oc_meta repository. The documentation covers the input structure, curation workflow, OMID assignment, RDF conversion, provenance and change tracking, dataset distributions, and access services.

5 Quality and security

5.1 Quality of the research output

5.1.1 Measures implemented to ensure quality

Before RDF conversion, the Curator performs identifier-based deduplication, enrichment, normalization, and error correction. Entities are merged only when shared identifiers establish that they represent the same resource. Provenance and change-tracking records preserve the history of each entity. META-CSV data can be validated before ingestion, and automated monitoring queries can detect database-level errors after ingestion.

5.2 Security measures

5.2.1 Storage

Other

Comment:

Curated metadata are stored as CSV and RDF files and in the OpenCitations Meta database. The current database distribution uses OpenLink Virtuoso, with provenance available both as RDF and through a dedicated queryable database.

5.2.2 Backup

Recovery is supported by versioned dataset dumps preserved in Zenodo and Figshare, which provide recoverable snapshots of published OpenCitations Meta releases.

Software for the ingestion process of OpenCitations Index

The OpenCitations Index software performs the final Index Process of the ingestion workflow. It converts citation links expressed with external identifiers into OMID-to-OMID relationships using OpenCitations Meta mappings, then generates OCDM-compliant citation entities identified by Open Citation Identifiers (OCIs).

Template: [Data Management Plan Template - University of Bologna](#)

Type: [Dataset](#)

1 Your research output

1.1 Contact details and responsibilities

1.1.1 Contact person(s)

Contact Name	Contact ID	E-mail address	Affiliation
Heibi, Ivan	0000-0001-5366-5194	ivan.heibi2@unibo.it	ALMA MATER STUDIORUM - Università di Bologna

1.1.3 Research output creator(s) and contributor(s)

Contributor Name	Contributor ID	E-mail address	Affiliation	Role
Moretti, Arianna	0000-0001-5486-7070	arianna.moretti 4@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator

Soricetti, Marta	0009-0008-1466-7742	marta.soricetti2@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator
Peroni, Silvio	0000-0003-0530-4305	silvio.peroni@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Other (project administration, resources and supervision)

1.1.4 Institution(s) involved

ALMA MATER STUDIORUM - Università di Bologna

1.2 Resources for data management

1.2.1 What resources will be dedicated to data management and ensuring that data will be FAIR?

Resources supporting the management and reuse of the Index ingestion software include the public OpenCitations GitHub repository, version control, source code, configuration files and tests

1.3 Identifying the research output

1.3.2 Which type of research output are you describing?

Software

2 Summary

2.3 Identification of software

2.3.1 Are you re-using already existing software?

Yes

2.3.2 Re-using existing software: why and how

oc-index is reused because it implements the Index Process that converts citation links based on external identifiers into OMID-based citation entities for the OpenCitations Index.

oc-index is released under the ISC License, which permits use, copying, modification, and distribution provided that the copyright and permission notice are retained.

2.4 Software characteristics

2.4.1 Software name

oc-index (OpenCitations Index software)

2.4.2 Type of software

Application software for citation data ingestion and transformation

2.4.3 Software formats

- a. Python source code (.py)
- b. TOML project configuration (.toml)
- c. INI configuration files (.ini)
- d. CSV lookup and configuration files (.csv)

2.4.4 Size of software

other

2.4.5 How is software created/used

oc-index is an open-source Python package used in the Index Process of the OpenCitations ingestion workflow. It receives citation tables produced during source preprocessing, in which citing and cited resources are represented by external persistent identifiers.

Using mappings generated by the Meta Process, the software resolves these identifiers to OpenCitations Meta Identifiers (OMIDs) and converts each citation into an OMID-to-OMID relationship. These relationships are used to generate citation entities for the OpenCitations Index.

Version 1.0.1 requires Python 3.10–3.12 and uses dependencies including oc-ocdm, oc-idmanager, RDFLib, pandas, Redis, and SPARQLWrapper. Configuration is provided through an INI file, and development history is maintained with Git.

2.4.6 Why is the software created/used

oc-index automates the integration of citation links from heterogeneous sources into the OpenCitations Index. By converting different external identifier schemes to OMIDs, it provides a common representation for identifying and deduplicating citations. The same workflow can be reused when new citation sources are added.

2.4.7 Software description

oc-index is a Python package for producing citation data for the OpenCitations Index. It resolves external identifiers in citation tables to OpenCitations Meta Identifiers (OMIDs), converts citation links into OMID-to-OMID relationships, and generates citation entities compliant with the OpenCitations Data Model. The software is open source and released under the ISC License.

3 Ethics and legal aspects

3.1 Personal data

3.1.1 Are you handling personal data?

No

3.2 Legal aspects

3.2.1 Are there any legal issues associated to your output? (e.g. IPR or valorization)

No

3.3 Other ethical aspects

3.3.1 Are there other ethical issues associated to your output?

No

4 Making research FAIR

4.1 Long-term preservation

4.1.1 Selecting what to preserve

Source code, version history, configuration files, documentation, tests, and released versions will be preserved. Earlier versions support the reproduction of historical ingestion workflows and OpenCitations Index releases.

4.1.2 Chosen repository/ies

Name	Link	Is your repository OpenAIRE compliant?	Is your repository catalogued in re3data?	What type of repository is it?
Software Heritage Archive	https://archive.softwareheritage.org/	No	Yes	Discipline specific

Comment:

Software Heritage provides an independent archive of the oc-index source code and development history, while GitHub remains the active development repository.

4.2 Findability

4.2.1 Persistent identifier(s)

Type of PID (e.g., DOI, Handle)	PID (e.g., doi.org/10...)
SWHID	<code>swh:1:snp:edecc78aff97644d83e48f7373710554a2f55721</code>

4.2.2 Metadata

The repository includes structured software metadata for the name, version, description, license, author contact, Python requirements, dependencies, and command-line entry points in pyproject.toml. GitHub records the development history, while Software Heritage records the archived software snapshot.

4.2.3 Keywords

OpenCitations, OpenCitations Index, citation ingestion, research software, Python, OMID, OCI, RDF, OpenCitations Data Model, Linked Open Data

4.3 Accessibility

4.3.1 Is the research output openly accessible?

Yes

4.3.3 Accessing the research output

The source code, documentation, configuration files, tests, and version history are publicly available in the OpenCitations GitHub repository at <https://github.com/opencitations/index>. The repository can be cloned with Git or downloaded directly, while archived source-code states are available through Software Heritage.

4.3.4 Is metadata openly accessible?

Yes

4.4 Interoperability

4.4.1 Methodologies

oc-index is developed and versioned with Git. In the ingestion workflow, Source Preprocess converts citation data into standardized CSV tables, while the Meta Process creates mappings between external persistent identifiers and OMIDs. oc-index uses these tables and mappings to generate OMID-to-OMID citation relationships and citation entities for the OpenCitations Index. Configuration is kept separate from the source code, and dependencies and command-line entry points are defined in the Python project configuration.

4.4.2 Vocabularies, taxonomies and other standards

oc-index produces citation data compliant with the OpenCitations Data Model (OCDM), represented in RDF using terms from the OpenCitations Ontology (OCO). Citation entities use CiTO, while OpenCitations Meta Identifiers (OMIDs) identify the citing and cited bibliographic resources and Open Citation Identifiers (OCIs) identify the citation entities. CSV is used as a tabular interchange format in the ingestion workflow, with RDF and Scholix used for published Index distributions.

4.5 Reusability

4.5.1 Licensing

other

Comment:

oc-index is released under the ISC License.

4.5.2 Documentation

The GitHub repository provides README.md for installation and requirements, USAGE.md for execution instructions, config.ini for configuration, and pyproject.toml for package metadata and dependencies. The OpenCitations Index paper documents the software's role within the Source Preprocess, Meta Process, and Index Process workflow.

5 Quality and security

5.1 Quality of the research output

5.1.1 Measures implemented to ensure quality

Development is managed with Git and supported by an automated test suite and GitHub workflows. Static type checking is performed with Pyright, and code changes can be reviewed through GitHub pull requests. Data produced by the ingestion workflow is also subject to validation and post-ingestion monitoring.

5.2 Security measures

5.2.1 Storage

Cloud Storage Services

Comment:

The active source code is stored and versioned in the public OpenCitations GitHub repository. Git records the development history, and write access to the maintained repository is controlled through GitHub permissions.

5.2.2 Backup

Recovery relies on distributed Git history

Software for the ingestion process of OpenCitations Meta

oc_meta is open-source software for ingesting and processing bibliographic metadata for OpenCitations Meta. It performs automated curation, including normalization, deduplication, enrichment, and error correction, assigns OpenCitations Meta Identifiers (OMIDs), and generates RDF compliant with the OpenCitations Data Model. It also records provenance and change-tracking information.

Template: [Data Management Plan Template - University of Bologna](#)

Type: [Dataset](#)

1 Your research output

1.1 Contact details and responsibilities

1.1.1 Contact person(s)

Contact Name	Contact ID	E-mail address	Affiliation
Massari, Arcangelo	0000-0002-8420-0696	arcangelo.massari@unibo.it	ALMA MATER STUDIORUM - Università di Bologna

1.1.3 Research output creator(s) and contributor(s)

Contributor Name	Contributor ID	E-mail address	Affiliation	Role
Mariani, Fabio	0000-0002-7382-0187	fabio.mariani@leuphana.de	Leuphana Universität Lüneburg	Creator

Persiani, Simone		simone.persiani2@studio.unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator
Peroni, Silvio	0000-0003-0530-4305	silvio.peroni@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Other (conceptualization)

1.1.4 Institution(s) involved

- a. ALMA MATER STUDIORUM - Università di Bologna
- b. Leuphana Universität Lüneburg

1.2 Resources for data management

1.2.1 What resources will be dedicated to data management and ensuring that data will be FAIR?

Resources supporting the management and reuse of oc_meta include its public GitHub repository, version control, source code, tests, software releases and the documented automated curation and RDF-generation workflow.

1.3 Identifying the research output

1.3.2 Which type of research output are you describing?

Software

2 Summary

2.3 Identification of software

2.3.1 Are you re-using already existing software?

Yes

2.3.2 Re-using existing software: why and how

oc_meta is reused because it implements the curation and RDF-generation workflow used to ingest and update bibliographic metadata in OpenCitations Meta.

oc_meta is released under the ISC License, which permits use, copying, modification, and distribution provided that the copyright and permission notice are retained.

2.4 Software characteristics

2.4.1 Software name

oc_meta (OpenCitations Meta software)

2.4.2 Type of software

Application software for bibliographic metadata ingestion, curation, and RDF generation

2.4.3 Software formats

- a. Python source code (.py)
- b. YAML configuration files (.yaml)
- c. TOML project configuration (.toml)
- d. CSV input and example files (.csv)
- e. JSON configuration and metadata files (.json)

2.4.4 Size of software

megabytes (MB)

Comment:

Version 3.0.0 is approximately 4.6 MB as a source distribution and 423 kB as a Python wheel, with a total release size of about 5.0 MB

2.4.5 How is software created/used

oc_meta is an open-source Python package that processes bibliographic metadata in structured CSV format. It normalizes values, identifies duplicate resources, enriches records, and corrects detectable errors before generating RDF compliant

with the OpenCitations Data Model and recording provenance and change-tracking information.

Version 2.1.3 supports Python 3.10–3.13 and uses OpenCitations components including oc-ds-converter and oc-ocdm, together with libraries such as pandas and RDFLib. Development history is maintained with Git.

2.4.6 Why is the software created/used

oc_meta automates the ingestion and curation of bibliographic metadata for OpenCitations Meta. It provides a repeatable workflow for processing heterogeneous metadata, managing OpenCitations Meta entities, generating OCDM-compliant RDF, and recording provenance as new bibliographic data are added.

2.4.7 Software description

oc_meta is open-source Python software for managing bibliographic metadata in OpenCitations Meta. It processes structured CSV metadata, performs automated curation, and generates RDF compliant with the OpenCitations Data Model. The software also records provenance and change tracking for bibliographic entities and is released under the ISC License.

3 Ethics and legal aspects

3.1 Personal data

3.1.1 Are you handling personal data?

Yes

3.1.2 Solutions adopted for handling personal data

oc_meta processes publicly available scholarly metadata from external bibliographic sources, including author and editor names and ORCID identifiers. It does not collect data directly from research participants, so the workflow does not involve participant recruitment or informed consent. Provenance records retain the source of imported metadata and subsequent changes.

3.2 Legal aspects

3.2.1 Are there any legal issues associated to your output? (e.g. IPR or valorization)

No

3.3 Other ethical aspects

3.3.1 Are there other ethical issues associated to your output?

No

4 Making research FAIR

4.1 Long-term preservation

4.1.1 Selecting what to preserve

Source code, released packages, version history, configuration files, documentation, tests, and software metadata will be preserved. Earlier releases are retained to support the reproducibility of previous OpenCitations Meta processing workflows.

4.1.2 Chosen repository/ies

Name	Link	Is your repository OpenAIRE compliant?	Is your repository catalogued in re3data?	What type of repository is it?
Zenodo	DOI: 10.5281/zenodo.20965287	Yes	Yes	Generalist

Comment:

Versioned oc_meta releases are preserved in Zenodo and independently archived in Software Heritage.

4.2 Findability

4.2.1 Persistent identifier(s)

Type of PID (e.g., DOI, Handle)	PID (e.g., doi.org/10...)
DOI	10.5281/zenodo.20965287

4.2.2 Metadata

oc_meta includes structured metadata for its title, creators, version, description, license, repository, release date, dependencies, and citation information. The source repository provides software metadata through files including CITATION.cff, pyproject.toml, and CHANGELOG.md, while Zenodo provides metadata for archived releases.

4.2.3 Keywords

OpenCitations, OpenCitations Meta, oc_meta, bibliographic metadata, Python, RDF, OMID, provenance, data curation, Semantic Web

4.3 Accessibility

4.3.1 Is the research output openly accessible?

Yes

4.3.3 Accessing the research output

The source code is publicly available at https://github.com/opencitations/oc_meta, and the software can be installed from PyPI. Versioned releases are available through Zenodo, archived source code through Software Heritage, and user and developer documentation through the OpenCitations documentation site.

4.3.4 Is metadata openly accessible?

Yes

4.4 Interoperability

4.4.1 Methodologies

oc_meta is developed and versioned with Git. Bibliographic metadata enter the workflow in structured CSV format, undergo automated curation, and are converted to RDF compliant with the OpenCitations Data Model. Configuration is kept separate from the source code, while dependencies and runtime requirements are declared in pyproject.toml. Automated tests are used during development.

4.4.2 Vocabularies, taxonomies and other standards

oc_meta generates RDF compliant with the OpenCitations Data Model. Provenance and change tracking follow the OpenCitations provenance model and PROV concepts. OMIDs identify entities managed by the software, while external identifiers include DOI, PMID, ISBN, ISSN, and ORCID. CSV is used as the tabular interchange format for bibliographic metadata.

4.5 Reusability

4.5.1 Licensing

other

Comment:

oc_meta is released under the ISC License.

4.5.2 Documentation

The GitHub repository and OpenCitations documentation site provide installation, configuration, usage, and development documentation, together with examples, citation metadata, and release information. The canonical OpenCitations Meta paper documents the underlying curation and RDF-generation workflow.

5 Quality and security

5.1 Quality of the research output

5.1.1 Measures implemented to ensure quality

oc_meta is maintained with Git and an automated test suite integrated with GitHub workflows. Development tools include pytest and coverage for testing, Pyright for static type checking, and Ruff for code analysis. Software changes are documented through versioned releases and the project changelog.

5.2 Security measures

5.2.1 Storage

Cloud Storage Services

Comment:

The active source code is stored and versioned in the public OpenCitations GitHub repository. Git records the development history, and write access to the maintained repository is controlled through GitHub permissions.

5.2.2 Backup

Recovery relies on distributed Git history, versioned Zenodo releases, and the independent Software Heritage archive, which provide recoverable copies of released software states.

Software for handling data compliant with the OpenCitations Data Model

oc_ocdm is an open-source Python library for creating and managing RDF data compliant with the OpenCitations Data Model (OCDM). It provides Python interfaces for bibliographic resources, agents, identifiers, provenance, and dataset metadata, and supports validation, change tracking, serialization, and synchronization with RDF stores.

Template: [Data Management Plan Template - University of Bologna](#)

Type: [Dataset](#)

1 Your research output

1.1 Contact details and responsibilities

1.1.1 Contact person(s)

Contact Name	Contact ID	E-mail address	Affiliation
Persiani, Simone		simone.persiani2@studio.unibo.it	ALMA MATER STUDIORUM - Università di Bologna

1.1.3 Research output creator(s) and contributor(s)

Contributor Name	Contributor ID	E-mail address	Affiliation	Role
Daquino, Marilena	0000-0002-1113-7550	marilena.daquino2@unibo.it	ALMA MATER STUDIORUM - Università di	Creator

			Bologna	
Peroni, Silvio	0000-0003-0530-4305	silvio.peroni@unibo.it	ALMA MATER STUDIORUM - Università di Bologna	Creator

1.1.4 Institution(s) involved

ALMA MATER STUDIORUM - Università di Bologna

1.2 Resources for data management

1.2.1 What resources will be dedicated to data management and ensuring that data will be FAIR?

Resources supporting the FAIR management of oc_ocdm include its public GitHub repository, documentation, version control, automated tests, dependency management and published software packages.

1.3 Identifying the research output

1.3.2 Which type of research output are you describing?

Software

2 Summary

2.3 Identification of software

2.3.1 Are you re-using already existing software?

Yes

2.3.2 Re-using existing software: why and how

oc_ocdm is reused as the common Python library for creating, importing, modifying, validating, and exporting OCDM-compliant RDF data. OpenCitations applications can use the library instead of implementing RDF manipulation, provenance, and serialization functions separately.

oc_ocdm is released under the ISC License, which permits use, copying, modification, and distribution provided that the copyright and permission notice are retained.

2.4 Software characteristics

2.4.1 Software name

oc_ocdm (software for handling OCDM-compliant data)

2.4.2 Type of software

Python software library for RDF object mapping, data manipulation, and provenance management

2.4.3 Software formats

- a. Python source code (.py)
- b. TOML project configuration (.toml)
- c. Markdown documentation (.md)
- d. Jupyter notebooks (.ipynb)

2.4.4 Size of software

kilobytes (KB)

Comment:

Version 11.0.22 is approximately 768 kB as a source distribution and 106 kB as a Python wheel.

2.4.5 How is software created/used

oc_ocdm is an open-source Python library built on RDFLib. It represents OCDM entities as Python objects and provides components for importing, modifying, validating, and exporting RDF data.

GraphSet manages bibliographic entities, ProvSet manages provenance snapshots, and MetadataSet manages dataset metadata. Reader imports RDF from files or triplestores, while Storer serializes data or synchronizes changes with RDF stores.

The library tracks changes to entities and generates provenance for creation, modification, merging, and deletion. Version 11.0.22 supports Python 3.10–3.13, and development history is maintained with Git.

2.4.6 Why is the software created/used

oc_ocdm simplifies the development of software that works with OCDM-compliant data. It lets Python developers create and manipulate bibliographic and citation data without implementing RDF, SPARQL, and ontology handling directly. The library is shared across OpenCitations applications and can also be reused by external projects that adopt OCDM.

2.4.7 Software description

oc_ocdm is an open-source Python library for creating and manipulating RDF data compliant with the OpenCitations Data Model. It provides object-oriented interfaces for bibliographic resources, agents, identifiers, provenance, and dataset metadata, with support for validation, change tracking, serialization, and synchronization with RDF stores. The software is released under the ISC License.

3 Ethics and legal aspects

3.1 Personal data

3.1.1 Are you handling personal data?

No

3.2 Legal aspects

3.2.1 Are there any legal issues associated to your output? (e.g. IPR or valorization)

No

3.3 Other ethical aspects

3.3.1 Are there other ethical issues associated to your output?

No

4 Making research FAIR

4.1 Long-term preservation

4.1.1 Selecting what to preserve

Source code, released packages, version history, tests, documentation, configuration files, and software metadata will be preserved. Earlier releases are retained because OpenCitations applications may depend on specific versions and because they support reproducibility.

4.1.2 Chosen repository/ies

Name	Link	Is your repository OpenAIRE compliant?	Is your repository catalogued in re3data?	What type of repository is it?
Zenodo	https://zenodo.org	Yes	Yes	Generalist

Comment:

Versioned oc_ocdm releases are preserved in Zenodo, while active development and version control remain in the OpenCitations GitHub repository.

4.2 Findability

4.2.1 Persistent identifier(s)

Type of PID (e.g., DOI, Handle)	PID (e.g., doi.org/10...)
DOI	10.5281/zenodo.5770646

4.2.2 Metadata

oc_ocdm includes structured metadata for its name, creators, version, license, Python requirements, repository, release information, and citation information. The source repository provides these through files including pyproject.toml, CITATION.cff, CHANGELOG.md, and README.md, while Zenodo and PyPI expose metadata for published releases.

4.2.3 Keywords

OpenCitations, oc_ocdm, OpenCitations Data Model, OCDM, Python, RDF, SPAR Ontologies, provenance, Semantic Web, object mapping

4.3 Accessibility

4.3.1 Is the research output openly accessible?

Yes

4.3.3 Accessing the research output

The source code is publicly available at https://github.com/opencitations/oc_ocdm, and the library can be installed from PyPI. Versioned releases are available through Zenodo, while documentation, API references, examples, and notebooks are provided through the OpenCitations documentation site and source repository.

4.3.4 Is metadata openly accessible?

Yes

4.4 Interoperability

4.4.1 Methodologies

oc_ocdm is developed and versioned with Git and follows a Test Driven Development approach. OCDM entities are represented as Python objects backed by RDF graphs, allowing data to be imported from RDF files or triplestores, modified through the Python API, tracked for changes, and serialized or synchronized with persistent RDF stores. The library is packaged and versioned independently from the applications that use it, so each project can depend on a specific release.

4.4.2 Vocabularies, taxonomies and other standards

oc_ocdm implements the OpenCitations Data Model and operates on RDF data using terms from the OpenCitations Ontology and the SPAR Ontologies. OCDM defines the representation of bibliographic resources, citations, agents, identifiers, and their relationships, while PROV-O is used for provenance. Supported RDF serializations include JSON-LD, N-Triples, and N-Quads, and changes can be expressed through SPARQL updates.

4.5 Reusability

4.5.1 Licensing

other

Comment:

oc_ocdm is released under the ISC License.

4.5.2 Documentation

The GitHub repository and OpenCitations documentation site provide the README, API and usage documentation, examples, notebooks, citation metadata, and changelog. The canonical oc_ocdm paper explains the library architecture and its main components, including Reader, Storer, CounterHandler, GraphSet, ProvSet, and MetadataSet, as well as the workflows for data import, manipulation, change tracking, provenance generation, and RDF synchronization.

5 Quality and security

5.1 Quality of the research output

5.1.1 Measures implemented to ensure quality

oc_ocdm uses Test Driven Development, automated tests, benchmarks, and GitHub workflows to check software changes. The library also validates entity operations against OCDM constraints. Change tracking and provenance record modifications to RDF entities, while versioned releases and the changelog document changes to the software.

5.2 Security measures

5.2.1 Storage

Cloud Storage Services

Comment:

The active source code is stored and versioned in the public OpenCitations GitHub repository. Git records the development history, and write access to the maintained repository is controlled through repository permissions.

5.2.2 Backup

Recovery relies on distributed Git history, published PyPI packages, and versioned Zenodo releases(<https://doi.org/10.5281/zenodo.5770646>), which provide

independent copies of released software states if the active repository becomes unavailable.

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